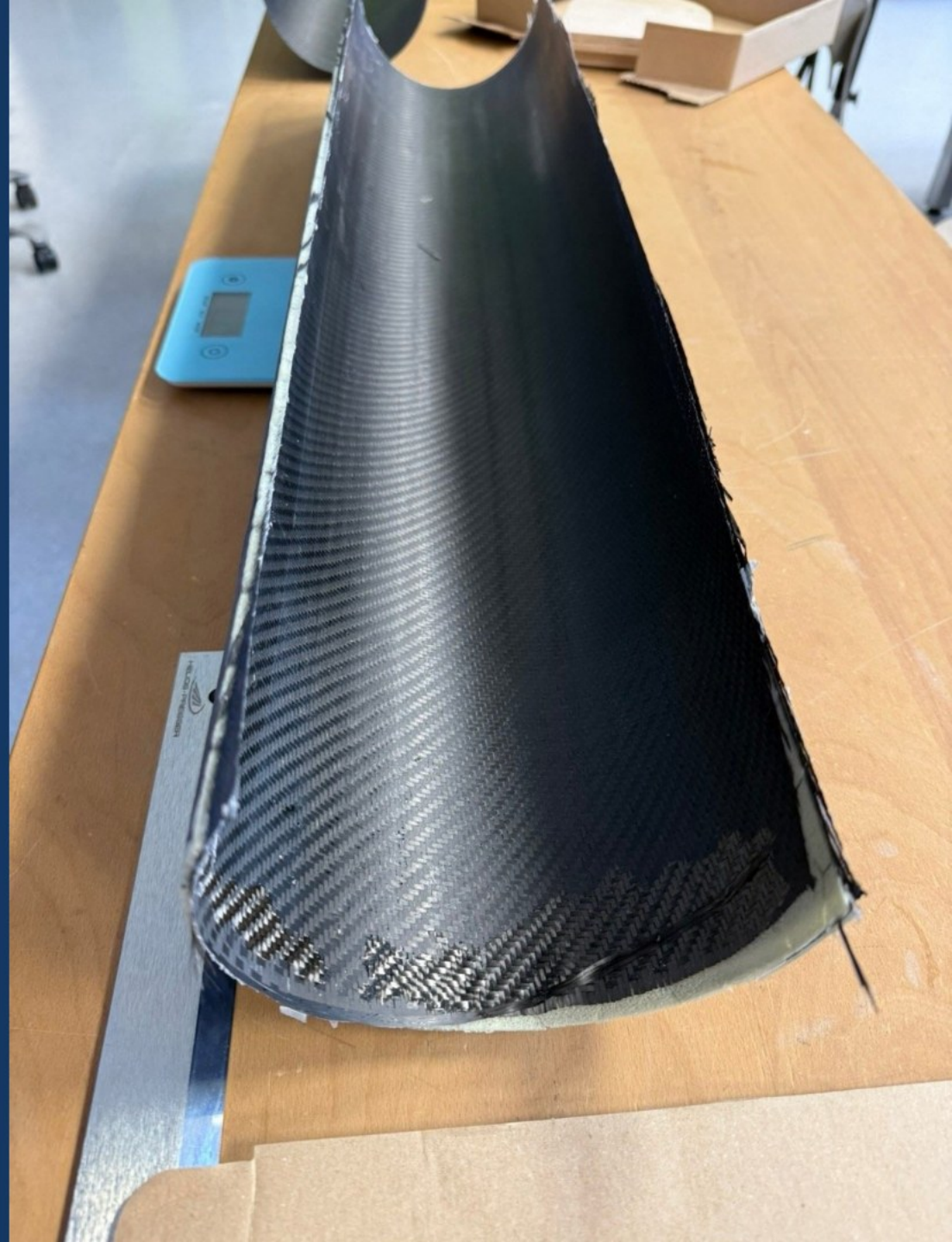
ERASMUS WORKSHOP · 18 NOVEMBER 2026

Build a carbon rotor blade

Sandwich construction with vacuum infusion

Windmobile · LTEtt



Eight carbon blades on one bicycle rim



WINDMOBILE

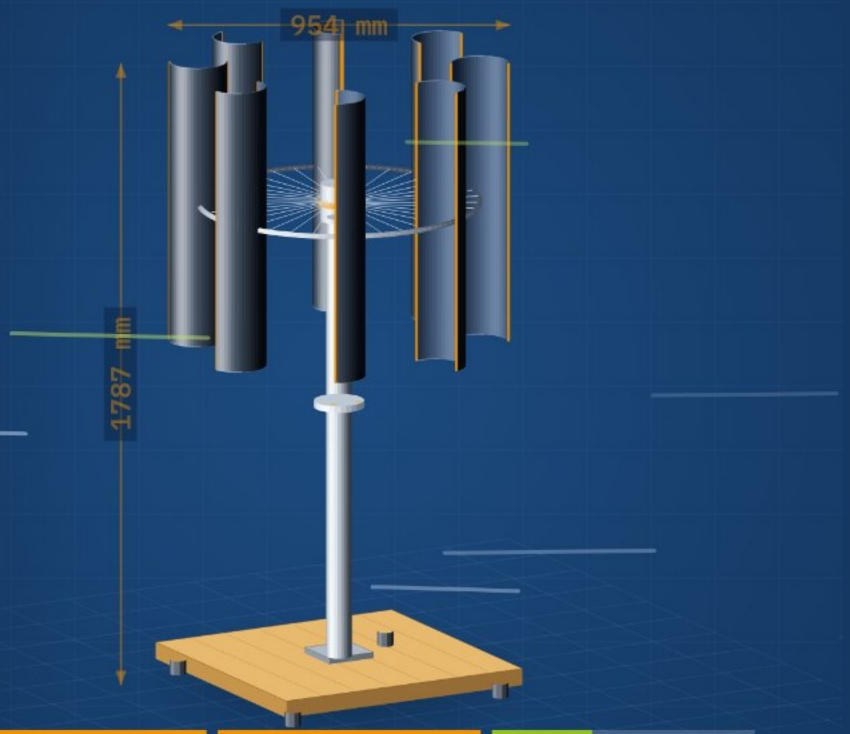
WIND

05 / 05

THE WIND DOES THE REST

Vertical-axis rotor

954 × 1787 mm



Five layers on a pipe

Layer stack

5. **Vacuum bag (tube film)**
   width 86 cm
4. **Peel ply + flow mesh**
   87 × 32 cm
3. **Carbon**
   85 × 30 cm
2. **Core: 3DCore PET 100, 3 mm**
   85 × 30 cm
1. **Carbon, visible side outside**
   85 × 30 cm

Process

1. **Tape and cut carbon, core, peel ply + flow mesh**
2. **Lay up the layers on the pipe**
3. **Slide into the vacuum bag, attach and seal the hoses, pull the vacuum**
4. **Mix the resin in 3 portions of 200 g + 60 g hardener, then infuse**
5. **Cure under the infrared lamp**

Step by step: prepare the materials



1 Tape and cut the carbon

2 pieces, 85 × 30 cm

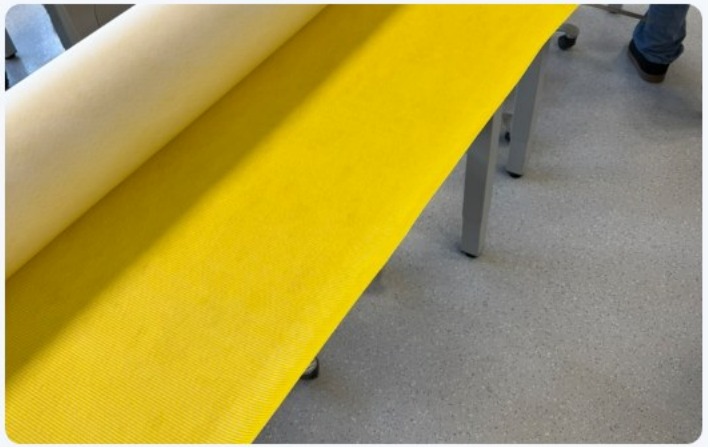
Tape the cutting lines so the fabric does not fray.



2 Cut the core

85 × 30 cm

3DCore PET 100, 3 mm, the same size as the carbon.



3 Cut peel ply + flow mesh

87 × 32 cm

One material from the roll, a little larger than the blade.

Step by step: lay up the layers



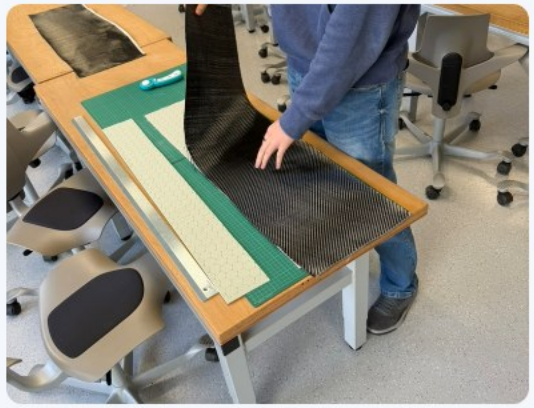
1 Carbon

Aligned and without wrinkles on the pipe. This side stays visible outside.



2 Core

Fix it with spray adhesive. It stiffens the blade and lets the resin flow.



3 Carbon

The second layer on top gives strength and stability.



4 Peel ply + flow mesh

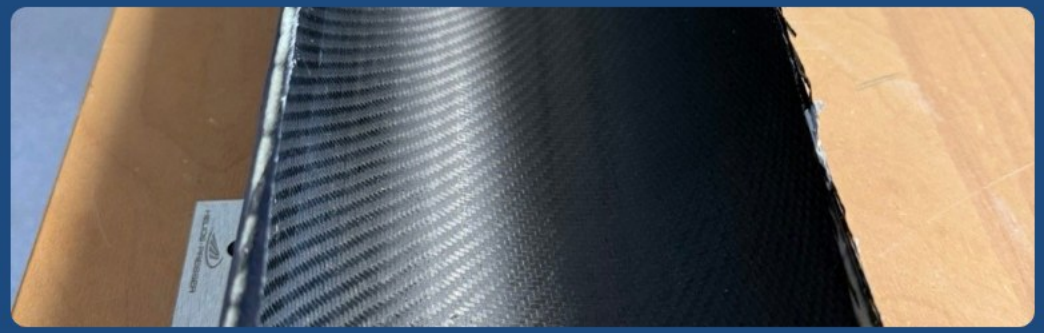
Spreads the resin. Pulled off after curing.

Step by step: bag, vacuum, infuse



1. Slide the lay-up into the vacuum bag
2. Attach and seal the resin hose and the vacuum hose
3. Connect the pump and pull the vacuum
4. Mix the resin in 3 portions: 200 g resin + 60 g hardener each
5. Open the resin hose and watch the flow front
6. Cure under the infrared lamp

The workshop blade and the finished blade



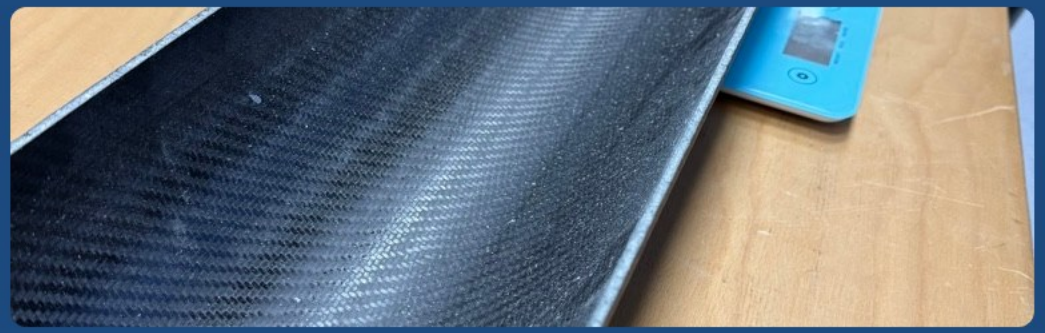
WORKSHOP RECIPE

634 g

Core: 3DCore PET 100, 3 mm

Resin 600 g + hardener 180 g

2 carbon layers



FINISHED BLADE

545 g

C profile Ø160 × 800 mm

Carbon – core – carbon

8 blades per turbine

Weigh your blade and compare it with ours.

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